$P_{Ar} = x_{Ar} P_{Total}$ $= 0.093 \times 2$

$= 0.186$ atm.

$P_{Xe} = x_{Xe} P_{Total}$ $= 0.312 \times 2$

$= 0.624$ atm.

Evaluate Yourself

6. (a) A mixture of He and O_2 were used in the 'air' tanks of underwater divers for deep dives. For a particular dive 12 dm^3 of O_2 at 1 *atm*. and 46 dm^3 of He, at 1 *atm*. were both pumped into a 5 dm^3 tank. Calculate the partial pressure of each gas and the total pressure in the tank at 298 K

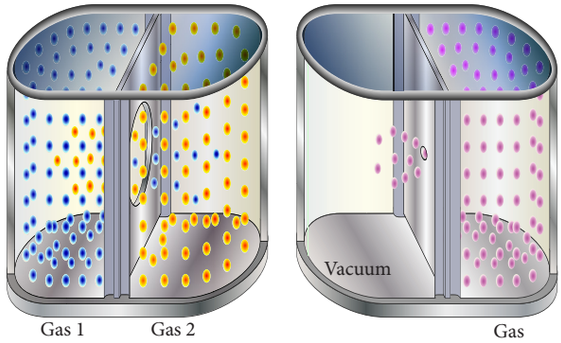
(b) A sample of solid $KClO_3$ (potassium chlorate) was heated in a test tube to obtain O_2 according to the reaction

$$2KClO_3 \rightarrow 2KCl + 3O_2$$

The oxygen gas was collected by downward displacement of water at 295 K. The total pressure of the mixture is 772 mm of Hg. The vapour pressure of water is 26.7 mm of Hg at 300K. What is the partial pressure of the oxygen gas?

6.4.1 Graham' s Law of Diffusion

Gases have a tendency to occupy all the available space. When two non -reactive gases are allowed to mix, the gas molecules migrate from region of higher concentration to a region of lower concentration. This property of gas which involves the movement of the gas molecules through another gases is called diffusion. Effusion is another process in which a gas escapes from a container through a very small hole.



Diffusion Effusion

Figure. 6.7 Diffusion and effusion of gases

The rate of diffusion or effusion is inversely proportional to the square root of molar mass. This statement is called Graham's law of diffusion/effusion.

$$\text{Mathematically, rate of diffusion } \alpha \frac{1}{\sqrt{M}}$$

Otherwise

$$\frac{r_A}{r_B} = \sqrt{\frac{M_B}{M_A}} \text{ -------- (6.15)}$$

When diffusing gases are at different pressures (P_A, P_B),

$$\frac{r_A}{r_B} = \frac{P_A}{P_B}\sqrt{\frac{M_B}{M_A}} \text{ ------(6.16)}$$

where r_A and r_B are the rates of diffusion of A and B and the M_A and M_B are their respective molar masses.



Graham's Law forms the basis of the process of enriching the isotopes of U^{235} from other isotopes.

Example:

1. An unknown gas diffuses at a rate of 0.5 time that of nitrogen at the same

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